

Verb position in German: Cycles and movement

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1 Introduction

At the beginning of HPSG research there were several linearization-based approaches to German constituent order (Reape 1994, Kathol 2000, Müller 1996, 1999, 2004, Wetta 2011, 2014). At the same time there were movement-based approaches which followed Borsley (1989) in assuming the analog of head-movement in HPSG. In grammars for German, the verb was assumed to be final and for V1 structures an initial version of the verb was related to the gap in final position (Kiss & Wesche 1991, Oliva 1992, Netter 1992, Frank 1994a, Kiss 1995a: Section 2.2.4.2, Feldhaus 1997: Section 3.1.1, Meurers 2000: Sections 4.1 and 5.1, Müller 2005a, 2023, Holler 2005: Section 6.2.5, Crysmann 2022).

The phenomenon of apparent multiple fronting (Müller 2003, 2005b, Bildhauer 2011) seems to show that verb movement is necessary for German (Müller 2023). In what follows, I discuss previous head-movement analyses and show their problems.

2 Previous analyses

Authors assuming head-movement either assume that head-movement is parallel to extraction and use nonlocal feature percolation to model the “movement” (Borsley 1989, Kiss & Wesche 1991, Kiss 1995a: Section 2.2.4.2) or they assumed that information is percolated along the head path (Oliva 1992, Meurers 2000: Sections 4.1 and 5.1). Meurers (2000) assumed a lexical rule taking a verb-final verb as input and licensing a verb-initial verb selecting for a fully saturated verbal projection containing a verb trace. The information about the verb trace is projected along the head path as the value of the feature DSL. (1) shows the lexical rule:

- (1) Rule for verb first placement (simplified):

$$\text{verb-initial-rule} \Rightarrow \left[\begin{array}{c} \left[\begin{array}{c} \text{SYNSEM|LOC|CAT} \\ \text{HEAD} \left[\begin{array}{c} \text{verb} \\ \text{VFORM } \textit{fin} \\ \text{INITIAL } + \\ \text{DSL } \textit{none} \end{array} \right] \\ \text{COMPS } \langle \left[\begin{array}{c} \text{LOC|CAT} \left[\begin{array}{c} \text{HEAD|DSL } \boxed{1} \\ \text{COMPS } \langle \rangle \end{array} \right] \rangle \end{array} \right] \rangle \\ \text{DTRS } \langle \left[\begin{array}{c} \text{SYNSEM|LOC } \boxed{1} \\ \text{CAT|HEAD} \left[\begin{array}{c} \text{verb} \\ \text{VFORM } \textit{fin} \\ \text{INITIAL } - \end{array} \right] \end{array} \right] \rangle \end{array} \right] \end{array} \right]$$

The verb trace is shown in (2):

- (2) Head trace according to Meurers (2000: 207):

$$\left[\begin{array}{c} \text{PHON } \langle \rangle \\ \text{SYNSEM|LOC } \boxed{1} \left[\text{CAT|HEAD|DSL } \boxed{1} \right] \end{array} \right]$$

Figure 1 shows the analysis of a the sentence in (3):

- (3) Kennt Aicke diese Band?
knows Aicke this band
‘Does Aicke know this band?’

The LOCAL value of the input of the lexical rule is the DSL value of the selected complement. Since DSL is a head feature, it is percolated down to the verb trace that shares its LOCAL value with its DSL value. This way the valence information and also information about the verb form is shared between the initial verb and the trace in final position.

Meurers's (2000) analysis has the problem that sequences like (4) are licensed as well:

- (4) *Kennt er das Buch kennt.
knows he the book knows

This is due to the fact that verbs cannot be specified as DSL *none* in the lexicon, since if they were, the verbs that are input to the V1 rule would be DSL *none* and the V1 verb would select something with the DSL value that corresponds to the LOCAL value of the input verb. If this is identified with the LOCAL value of the verb trace the cycle causes a contradiction. The value of SYNSEM|LOC|CAT|HEAD|DSL|LOC|CAT|HEAD|DSL is not compatible with *none*. Meurers suggests instead to restrict the DSL value on the fly. So he suggests the following constraint on finite verbal projections with a non-empty phonology list:

- (5) Constraint on DSL for overt verbs adapted from Meurers (2000: 207):

$$\left[\text{HEAD-DTR} \begin{bmatrix} \text{PHON} & \text{non-empty-list} \\ \text{SYNSEM} & [\text{LOC|CAT|HEAD|VFORM FIN}] \end{bmatrix} \right] \Rightarrow [\text{HEAD-DTR|SYNSEM|LOC|CAT|HEAD|DSL none}]$$

For example, this constraint applies to *kennt* in *das Buch kennt* and forces the DSL value to be *none*. This works well for examples like (4), but it fails for the grammatical sentence in (3). The reason is that there is a projection of the verb trace and the NP *diese Band* 'this band' that has both a non-empty phonology and a filled DSL value. Müller (2005a) made the constraint more specific and required the head daughter to be of type *word*:

- (6) Constraint on overt verbs by Müller (2005a: 17):

$$\left[\text{HEAD-DTR} \begin{bmatrix} \text{word} \\ \text{PHON non-empty-list} \end{bmatrix} \right] \Rightarrow [\text{SYNSEM|LOC|CAT|HEAD|DSL none}]$$

Müller (2005a: 13) suggested that the verb movement rule should be a syntactic rule rather than a lexical rule, since it then can apply to coordinated verbs like *kennt und liebt* 'knows and loves'.

- (7) Aicke kennt und liebt diese Band.
Aicke knows and loves this band

The problem is that the implication in (6) does not apply in cases like (8a):

- (8) a. *Gibt_j [der Affe [[lacht] und [dem Kind den Stock _{-j}]]].
gives the monkey laughs and the child the stick
b. dass [der Affe [[lacht] und [dem Kind den Stock gibt]]]
that the monkey laughs and the child the stick gives
'that the monkey laughs and gives the child the stick'

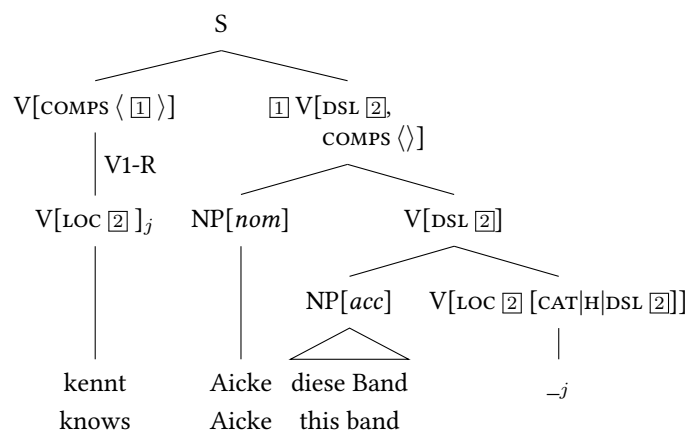
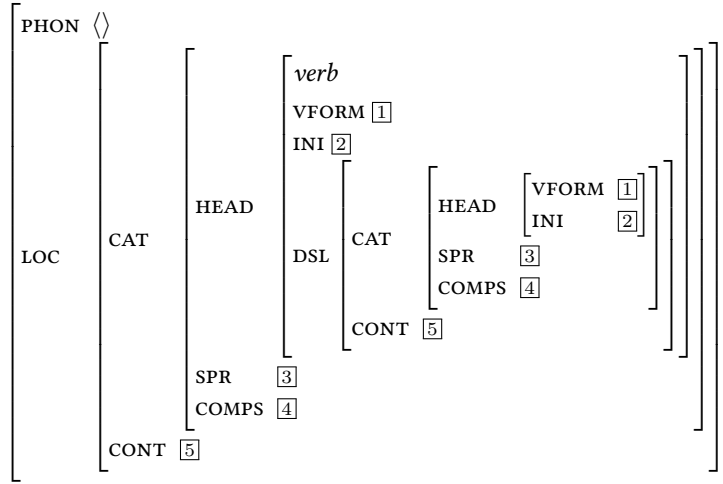


Figure 1: Analysis of *Kennt Aicke diese Band*.

(10) Non-cyclic verb-trace with explicit sharing of features:



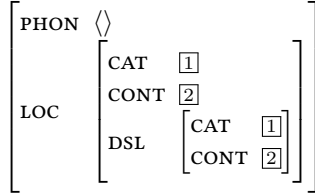
Since there is no cycle in DSL, all overt verb-final verbs can have the DSL value *none*.

A similar trace was provided by Crysmann (2022: 57), but Crysmann’s trace did suggest that the VFORM value of the trace is finite and it was not shared with the VFORM value inside of DSL. In the analysis of multiple fronting the verb trace can have other verb forms than finite (Müller 2005b: 320). Crysmann’s approach would not have made any wrong predictions, since the finite verb trace can function as trace for verbs in base or infinitive form. This is so since traces for different verbs are all just empty. A trace with VFORM finite and a corresponding element with VFORM *bse* would nevertheless be unintuitive.

As is obvious from (10), non-cyclicity requires the sharing of five different feature-value combinations in comparison to one structure sharing with the cyclic structure in (2). There is a missing generalization.

Apart from this, the problem of the distribution of the DSL value of a coordination to the conjuncts that was demonstrated in Figure 2 is not solved. A way to solve it, would be to locate DSL outside of CAT, which is shared among conjuncts (Pollard & Sag 1994: 202). The corresponding head trace is shown in (11):

(11) Head trace under LOCAL:



(11) needs less structure sharings than (10) since CAT and CONT can be shared. DSL is under local, so it is among the features that are shared in nonlocal dependencies, which is important for apparent multiple frontings as the one in (9). The disadvantage of (11) is that the percolation of DSL does not follow from anything. While it is taken care of if DSL is a head feature or is living under NONLOC, the percolation needs to be stipulated when the DSL feature is located under LOCAL.

3 Conclusion

It has been shown that verb movement as assumed for the Germanic V2 languages interacts in interesting ways with other phenomena. Coordination suggests that cyclic structures for the head trace are at the heart of the problem. An alternative way out of the situation could be the development of a new theory of coordination that takes care of the cases discussed here (see Abeillé & Chaves 2024 for an overview of approaches to coordination in HPSG). For the time being the non-cyclic verb-movement trace in (10) or a non-complete sharing of LOCAL values in the V1 rule seems to be the best option.

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